Olist E-Commerce | Big Data in Industry

Take-Home Assignment 1 - Business Performance Review

Dataset: Brazilian E-Commerce Public Dataset by Olist | Database: PostgreSQL | Pipeline: Python ELT

1. Project objective

Build a reproducible e-commerce analytics pipeline and deliver a business review covering data quality, sales, customer behavior and delivery performance. The workflow ingests nine Olist CSV files, transforms them into clean staging tables and creates an order-level analytical mart.

2. ELT architecture

Layer	Implementation	Purpose
Extract / Load	Python, pandas, SQLAlchemy; PostgreSQL raw schema	Load nine source CSVs as text; repeatable refresh
Transform	SQL in staging schema	Clean identifiers, cast dates and numeric values
Analytics mart	mart.order_summary	One row per order with product, freight, payment, review and delivery fields
Quality gate	Five SQL checks called from Python	Stop on critical errors; log payment mismatches for review
Outputs	SQL queries, Matplotlib charts, report	Support weekly business decisions

3. Operational approach

The pipeline reads database settings from a local .env file, checks for missing CSV files, writes status and errors to pipeline.log, and executes SQL transformations within a PostgreSQL transaction. Raw tables are replaced rather than appended to support reruns. The user verified an order-level mart containing 99,441 orders.

Reproducibility note: Final repository should include a README, requirements.txt, SQL files, example environment settings and instructions for downloading the Olist dataset. Do not commit passwords or raw data.

4. Scope and interpretation

Delivered GMV is the sum of product prices for delivered orders, excluding freight. Some other order-level analysis uses product price plus freight; these are separate measures and should not be presented as identical revenue definitions.

Part B | Data Quality Findings

Five SQL checks were executed in PostgreSQL and integrated into the ELT workflow. Results below reflect the user-verified query outputs.

Check	Observed result	Handling policy
Null or duplicate order IDs	0 missing; 0 duplicates	Critical: halt pipeline and investigate
Delivery before purchase	0 invalid dates	Critical: halt pipeline and investigate
Payment vs item + freight	303 mismatched orders	Warning: flag for reconciliation; do not silently delete
Missing customer references	0 missing customers	Critical: halt pipeline and investigate
Review score outside 1-5	0 invalid scores	Critical: halt pipeline and investigate

Quality policy and interpretation

Critical failures: Null/duplicate order identifiers, impossible delivery chronology, missing customer references and invalid review scores indicate structural or semantic problems. The Python quality-check function raises an exception to stop downstream mart creation when these counts are nonzero.

Payment reconciliation: The 303 differences are exceptions to investigate, not confirmed fraudulent or incorrect transactions. Possible causes such as vouchers or adjustments have not been verified from the provided results. These orders remain in the analytical dataset and are logged as warnings.

Failure handling: A simulated ValueError test confirmed the pipeline stops and logs the exception. This test does not itself prove that deliberately corrupted source data is caught; the SQL validation logic is the mechanism for detecting actual defects.

Recommended follow-up

Inspect a sample of the 303 flagged orders, compare payment methods and sequential payment rows, verify any applicable voucher rules, and document whether each difference is legitimate or requires correction. Preserve the original values and an audit trail.

Part C & D | SQL Insights and Metrics

Five analytical SQL queries

Query file	Technique	Business question
query1.sql	CTE + LAG	Monthly delivered order value and month-over-month growth
query2.sql	CTE + DENSE_RANK	Top product categories by product sales value
query3.sql	CTE + rolling AVG	Daily order value and seven observed-day moving average
query4.sql	CTE + window percentage	One-time vs repeat customers
query5.sql	CTE + DENSE_RANK	Late-delivery rate and average delay by customer state

Verified performance indicators

Indicator	Value	Definition
Delivered orders	96,478	Orders marked delivered
Delivered GMV	R$13,221,498.11	Delivered product price, excluding freight
Average review score	4.09 / 5	Average review score in the order-level mart
Late-delivery rate	8.11%	Delivered orders with known dates, arriving after estimate
Repeat-purchase share	3.00%	2,801 of 93,358 customers with delivered orders
One-time customer share	97.00%	90,557 of 93,358 customers with delivered orders

Top categories by product revenue

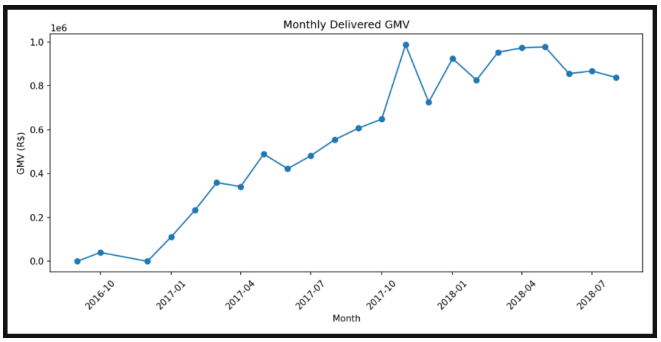
Rank	Category	Product revenue (R$)
1	health_beauty	1,233,131.72
2	watches_gifts	1,166,176.98
3	bed_bath_table	1,023,434.76
4	sports_leisure	954,852.55
5	computers_accessories	888,724.61

North Star: Monthly delivered GMV. Supporting measures: monthly growth, repeat-purchase share, late-delivery rate and review score. Repeat-purchase share describes the observed dataset; it is not a cohort retention rate.

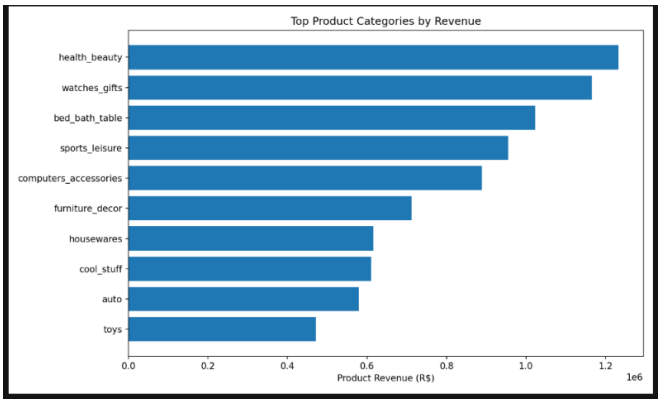
Part D | Business Visualizations

Four charts generated from saved SQL analyses and PostgreSQL data.

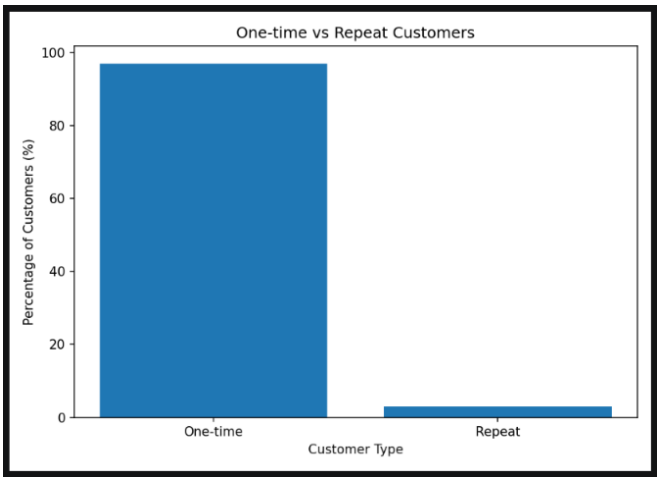
Monthly Delivered GMV



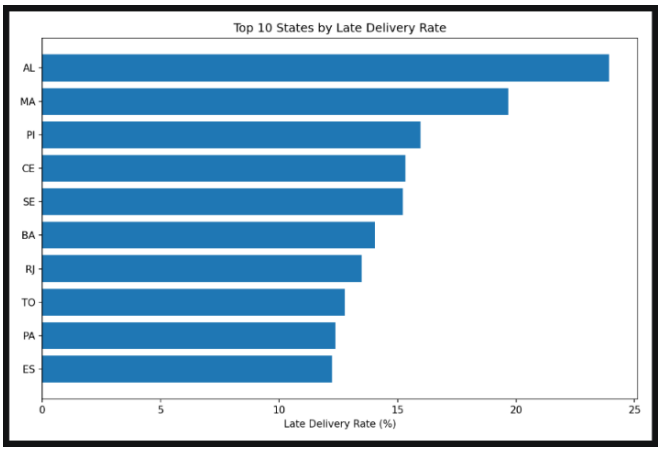
Top Product Categories



One-time vs Repeat Customers



Late Delivery by State



Interpretation

The GMV trend rises markedly through 2017 before fluctuating in 2018. Health & beauty, watches & gifts and bed & bath are the largest product categories shown. Repeat purchasing is limited to 3% of observed delivered-order customers. State-level late-delivery rates vary widely; AL is highest among the ten states displayed.

One-Page Memo | Vice President

To: Vice President, E-Commerce Operations
From: Business Data Analytics Team
Subject: Sales Performance, Retention and Delivery Reliability
Source: Olist Brazilian E-Commerce Dataset

Executive summary

Olist recorded **96,478 delivered orders** with **R$13.22 million in delivered product GMV** (excluding freight). Average review score was **4.09/5**. The largest opportunities appear in repeat purchasing, where only **3%** of observed customers ordered more than once, and logistics, where **8.11%** of delivered orders with known dates arrived late.

Key findings

Sales: Health & beauty (R$1.23M), watches & gifts (R$1.17M), and bed & bath (R$1.02M) were the leading product categories by product revenue. Monthly delivered GMV shows a strong increase through 2017, followed by fluctuations in 2018.

Customers: Of 93,358 customers with delivered orders, 90,557 purchased once and 2,801 purchased more than once. The observed 3% repeat-purchase share signals a potential growth opportunity, but the data alone does not establish why customers did not return.

Delivery: The overall late-delivery rate was 8.11%. Regional variation is substantial, with AL appearing highest in the top-ten state chart. More detailed volume-weighted investigation is needed before prioritizing operational interventions.

Data quality: Four critical checks returned zero violations. A separate payment-to-order-value comparison flagged 303 differences; these should be reconciled rather than automatically classified as errors.

Recommended actions

1. Improve delivery reliability. Review the states with the highest late rates, segment by order volume and logistics partner, and improve delivery-time estimates. Track late-delivery rate monthly.

2. Test customer re-engagement. Pilot post-purchase communication and targeted offers for first-time customers. Measure repeat-purchase share and compare with a control group.

3. Protect category growth. Work with leading sellers in health & beauty, watches & gifts, and bed & bath on stock availability and promotional effectiveness.

4. Reconcile payments. Investigate the 303 mismatches, document legitimate adjustments, and retain automated monitoring.

Decision requested

Approve a focused delivery-performance review and a measured customer re-engagement pilot. Use **monthly delivered GMV** as the North Star metric, supported by repeat-purchase share, late-delivery rate and customer review score.